

GO_Biological_Process_2023 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Steroid Metabolic Process (GO:0008202)	0.15944633	56	3.767e-05	2.022e-01	HSD17B2:14 HSD11B1:107 CYP11A1:201 CYP51A1:375 HSD17B3:409 CYP21A2:412
'De Novo' AMP Biosynthetic Process (GO:0	-0.15293445	4	2.895e-01	1.580e-01	PFAS:128 PAICS:5819 ATIC:5819 ADSS1:5819 NA NA
'De Novo' Post-Translational Protein Fol	0.09781599	15	1.898e-01	1.950e-01	SDF2L1:887 DNAJB13:1097 HSDPA14:1742 GAK:1896 ERO1A:1540 ENTPD5:6847
2-Oxoglutarate Metabolic Processes (GO:000	0.15259031	13	5.685e-02	1.950e-01	GOT2:166 TAT:169 OGDHL:185 GOT1:484 L2DHGDH:1540 IHDH:1683
3'-UTR-mediated mRNA Destabilization (GO	-0.02404704	11	1.825e-01	1.950e-01	DND1:1427 TRIM71:5819 RCHN1:5819 ZFP36L1:5819 ZFP36L2:5819 MOV10:5819
3'-UTR-mediated mRNA Stabilization (GO:0	0.07422310	11	1.057e-01	1.950e-01	MAPK14:1303 HNRNP13C:409 HNRNP10:1754 DAZL:2202 ZFP36L2:5819 TARDBP:6847
3'-Phosphoadenosine 5'-Phosphosulfate Me	0.24757905	8	1.532e-02	1.950e-01	SULT1A2:97 SULT1B1:964 SULT2B1:1030 SULT1A3:1405 SULT1E1:1680 PAPSS2:6847
5S Class rRNA Transcription By RNA Polym	-0.04055290	5	7.535e-01	1.950e-01	GTF3C2:5819 GTF3C3:5819 GTF3C4:5819 GTF3C5:5819 GTF3C6:5819 NA
7-Methylguanosine RNA Capping (GO:000945	0.20867202	2	3.067e-01	1.950e-01	RNMT:569 CMTR2:6847 NA NA NA NA
Methylguanosine mRNA Capping (GO:000663	0.20867202	2	3.067e-01	1.950e-01	RNMT:569 CMTR2:6847 NA NA NA NA
ADP Transport (GO:0015866)	-0.02679134	5	1.356e-01	1.950e-01	SLC25A24:470 SLC25A42:5819 SLC25A5:5819 SLC25A23:5819 SLC25A23:5819
AMP Biosynthetic Process (GO:0006167)	-0.18086332	6	1.250e-01	1.950e-01	PFAS:128 APRT:855 PAICS:5819 ATIC:5819 ADSS1:5819 NUDT12:5819
ATF6-mediated Unfolded Protein Response	0.05763797	5	6.554e-01	1.950e-01	CREBZF:656 MBTPS1:6847 XBP1:6847 ATF6B:6847 DDIT3:6847 NA
ATP Biosynthetic Process (GO:0006754)	0.02728418	13	7.335e-01	1.950e-01	TGFB1:780 ATP5F1C:1803 NUDT2:6847 VPS90I:6847 ATP5PF:6847 ATP5PD:6847
ATP Metabolic Process (GO:0046034)	0.02856910	25	6.212e-01	1.950e-01	TGFB1:780 ENPP3:1053 BAD:1081 MYH6:1785 ATP5F1C:1803 PARG:2089
ATP Synthesis Coupled Electron Transport	0.11562172	3	4.879e-01	1.950e-01	NDUFB:949 NDUFS2:6847 NDUUUF:5819
ATP Transport (GO:0015867)	-0.08235427	9	3.923e-01	1.950e-01	CALHM1:191 SLC25A24:470 SLC25A42:5819 SLC25A5:5819 SLC25A23:5819
AV Node Cell To Bundle Of His Cell Commu	0.04574441	4	7.514e-01	1.950e-01	GJA5:2469 SCN5A:6847 CXADR:6847 GJC3:6847 NA NA
Arp2/3 Complex-Mediated Actin Nucleation	-0.09664430	10	2.900e-01	1.950e-01	WHAMM:727 ARPC5L:796 WASHC1:1166 ARPC1A:1319 ARPC5:5819 ARPC3:5819
B Cell Activation (GO:0042113)	0.06407234	61	8.398e-02	1.950e-01	CD86:34 HDAC9:214 IFNK:316 IL4:679 CR2:1071 VCAN:1120 PTFN2:101
B Cell Chemotaxis (GO:0035754)	0.05649964	5	6.617e-01	1.950e-01	CYP7B1:734 PIK3CD:6847 HSD3B7:6847 CXCL13:6847 GAS6:6847 NA
B Cell Differentiation (GO:0030183)	0.05405264	44	2.153e-01	1.950e-01	HDAC9:214 IFNK:316 IL4:679 CR2:1071 VCAN:1120 PTFN2:101
B Cell Homeostasis (GO:0001782)	0.03948662	7	7.426e-01	1.950e-01	TNFRSF13B:400 MEF2C:1121 GAPT:6847 DOCK10:6847 NCKAP1:6847 TNFAIP3:6847
B Cell Proliferation (GO:0042100)	0.07943161	15	2.870e-01	1.950e-01	IFNK:316 CR2:1071 MEF2C:1121 PTPRC:2183 IL10:6847 CD70:6847
B Cell Receptor Signaling Pathway (GO:00	0.07446688	26	1.890e-01	1.950e-01	SH2B2:544 CTLA4:485 CD30:1042 MEF2C:1121 RFTN1:2025 PTPRC:2183
BMP Signaling Pathway (GO:0030509)	-0.04136061	42	3.542e-01	1.950e-01	LEFTY2:341 LEFTY1:489 GDF10:1102 FAM83G:1081 INHBE:1119 SLC39A5:1205
C-terminal Protein Amino Acid Modificat	-0.08813923	8	3.880e-01	1.950e-01	AGBL4:781 GLOP1:967 AGLB5:5819 AGTPBP1:5819 FOLH1:5819 ATG16L1:5819
C-terminal Protein Deglutamylation (GO:0	-0.13997713	4	3.323e-01	1.950e-01	AGBL4:781 AGLB5:5819 AGTPBP1:5819 FOLH1:5819 NA NA
C-terminal Protein Lipidation (GO:000650	-0.03478644	3	8.437e-01	1.950e-01	GLOP1:967 ATG16L1:5819 ATG7:10875 NA NA NA
C4-dicarboxylate Transport (GO:0015740)	0.11620399	10	2.030e-01	1.950e-01	SLC1A1:216 SLC25A12:6847 SLC25A18:992 SLC13A2:1451 SLC25A13:6847 SLC13A5:6847
CD4-positive, Alpha-Beta T Cell Activatio	-0.13020632	5	3.133e-01	1.950e-01	NGK7:147 STOML2:5819 TNFSF4:5819 RASD2:5819 HMGBL:5819
CD40 Signaling Pathway (GO:0023035)	0.03632786	7	7.393e-01	1.950e-01	CD86:34 CD40LG:6847 ITGB1:6847 TNF2:6847 TNF2:6847 ITGA5:6847
CENP-A Containing Chromatin Assembly (GO	0.04334790	5	5.737e-01	1.950e-01	NASP:1553 MSL18A:6847 CENPI:6847 HURP:6847 OIP5:6847 NA
COP1 Coating Of Golgi Vesicle (GO:004820	-0.05684820	2	7.807e-01	1.950e-01	ARFGAP3:1036 GBF1:10202 NA NA NA NA
COP1-coated Vesicle Budding (GO:0035964)	-0.05684820	2	7.807e-01	1.950e-01	ARFGAP3:1036 GBF1:10202 NA NA NA NA
COP1 Vesicle Coating (GO:0048208)	-0.03484854	14	6.561e-01	1.950e-01	KLHL12:475 TRAPPC12:1193 TRAPP3:5819 TRAPP2L:5819 TRAPP3:5819 TRAPP2C:5819
COP1-coated Vesicle Budding (GO:0090114	-0.01150596	29	8.303e-01	1.950e-01	TBC1D20:311 KLHL12:475 SH2D2A:491 TRAPP3:12193 TRAPP3:5819 TRAPP2L:5819
CRD-mediated mRNA Stabilization (GO:0070	-0.19756073	5	1.261e-01	1.950e-01	IGFBP1:496 YBX1:1195 DHX9:5819 PAPBC1:5819 HNRNPD:5819 NA
Cdc42 Protein Signal Transduction (GO:00	-0.04055702	6	7.308e-01	1.950e-01	SHTN1:5819 WAS:5819 RHCV:5819 NTN1:5819 RHOBT1B:5819 RHOBT2:5819
DNA Alkylation (GO:0006305)	0.09262387	15	2.144e-01	1.950e-01	DNMT3B:31 GATAD2A:146 BAZ2A:343 HELLS:1649 KMT2E:6847 MTRR:6847

EnrichmentHsSymbolsFile2 Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
NIKOLSKY_BREAST_CANCER_IQ21_AMPLICON	-0.29109711	27	1.661e-07	1.075e-03	ADAM15:45 PBXIP1:80 MTX1:85 RUSC1:89 RORC:177 FDPS:188
NIKOLSKY_BREAST_CANCER_Q012_Q13_AMPLICO	-0.14502570	100	5.637e-07	1.825e-03	COL20A1:30 HELZ28 MTG2:97 CDH26:104 FAM217B:130 LAMFAS:149
WP_METAPATHWAY_BIOTRANSFORMATION_PHASE_I	0.13697150	92	5.790e-06	1.250e-02	SULT1A2:97 AKR1D1:117 MGST1:161 CYP2F1:176 COMT:179 CYP11A1:201
KEGG_STEROID_HORMONE_BIOSYNTHESIS	0.23916671	26	2.444e-05	3.957e-02	HSD17B2:14 HSD11B1:107 AKR1D1:117 COMT:179 CYP11A1:201 HSD17B3:409
REACTOME_BIOLOGICAL_OXIDATIONS	0.09916207	123	1.503e-04	1.947e-01	ALDH1A1:73 SULT1A2:97 HPGDS:136 MGST1:161 CYP2F1:176 COMT:179
REACTOME_METABOLISM_OF_STEROID_HORMONES	0.19383617	28	3.868e-04	2.784e-01	HSD17B2:14 HSD11B1:107 CYP11A1:201 HSD17B3:409 CYP21A2:412 SERPINIA6:585
REACTOME_TRANSPORT_OF_INORGANIC_CATIONS_	0.10671382	95	3.319e-04	2.784e-01	SLC36A1:71 SLC36A2:135 SLC12A3:193 SLC1A1:216 SLC6A6:251 SLC7A6:272
REACTOME_METABOLISM_OF_LIPIDS	0.04463701	576	2.908e-04	2.784e-01	HSD17B2:14 SLC01B1:82 PTGR1:96 HSD11B1:107 AKR1D1:117 HPGDS:136
CAIRO_HEPATOBLASTOMA_DN	0.0740518	196	3.441e-04	2.784e-01	TMEM131L:35 AKR1D1:117 SLC46A3:125 CA2:152 SLC27A5:177 OGDHL:185
KEGG_CYTOKINE_CYTOKINE_RECEPTOR_INTERACT	0.07628784	180	4.324e-04	2.800e-01	IFNAR1:58 IL1R2:66 TNF:87 TNFSF10:103 TNFSF15:112 IL1RAP:304
NIKOLSKY_BREAST_CANCER_Q021_Q24_AMPLICON	-0.09226699	111	7.998e-04	3.176e-01	PLEC:24 FAM83B:220 GPR20:254 TRAPP2C:5819 RPLAHD:293 SCRIB:348
REACTOME_ERYTHROCYTES_TAKE_UP_OXYGEN_AND	0.43146604	5	8.338e-04	3.176e-01	CA2:152 CA4:253 AQP1:762 SLC4A1:1060 RHAG:2146 NA
REACTOME_SLC_MEDIATED_TRANSMEMBRANE_TRAN	0.06778672	219	5.733e-04	3.176e-01	SLC22A16:3 SLC36A1:71 SLC01B1:82 SLC10A6:83 SLC36A2:135 SLC12A3:193
WP_Q0112_COPY_NUMBER_VARIATION_SYNDROME	0.17941272	30	6.742e-04	3.176e-01	CNNM4:163 ARID5A:424 CNNM3:508 KANS1:580 COL2A1:605 DUSP2:677
WP_STEROID_HORMONE_PRECURSOR_BIOSYNTHESI	0.36775507	7	7.532e-04	3.176e-01	AKR1D1:117 CYP11A1:201 CYP21A2:412 CYP17A1:883 HSD3B2:1180 SRD5A2:2125
KEGG_ALANINE_ASPARTATE_AND_GLUTAMATE_MET	0.20122634	24	6.854e-04	3.176e-01	GOT2:166 AGXT:186 CAD:245 IL4I1:373 GLS2:427 GOT1:484
KEGG_CYSTINE_AND_METHIONINE_METABOLISM	0.2282245	21	7.602e-04	3.176e-01	DNMT3B:31 DNMT1:113 GOT2:166 TAT:169 IL4I1:373 LDHC:408
NIKOLSKY_BREAST_CANCER_17Q21_Q25_AMPLICO	-0.06055546	248	1.075e-03	3.783e-01	FTS3:58 HOXB13:67 LRRC59:71 ITGB4:100 ENGASE:119 EVPL:121
WP_CLASSICAL_PATHWAY_OF_STEROIDOGENESIS_	0.27189643	12	1.110e-03	3.783e-01	HSD11B1:107 CYP11A1:201 HSD17B3:409 CYP21A2:412 CYP17A1:883 HSD3B2:1180
BENPORATH_ES_WITH_H3K27ME3	-0.03573796	730	1.177e-03	3.810e-01	CEMP10 BNC1:14 PLEC:24 SUSD4:34 ADAM15:45 CAMK2B:46
NIKOLSKY_BREAST_CANCER_16Q24_AMPLICON	-0.14706730	40	1.297e-03	3.998e-01	PIEZO1:56 CPNE7:138 ZC3H18:408 ANKRD10:431 SPATA2:545 MVD:604
BENPORATH_SUZ12_TARGETS	-0.03630485	681	1.422e-03	4.187e-01	CEMP10 BNC1:14 PLEC:24 SUSD4:34 ADAMTSL3:50 HXQB1:367
REACTOME_GLUCCOARTICOID_BIOSYNTHESIS	0.36870547	6	1.762e-03	4.226e-01	HSD11B1:107 CYP21A2:412 SERPINIA6:585 CYP17A1:883 HSD3B2:1180 HSD11B2:6047
WP_ALLOGRAFT_REJECTION	0.11271893	65	1.691e-03	4.226e-01	CD86:34 TNF:87 HLA-DOB:137 FOXP3:324 ABCB1:397 IL1A:608
WP_PROXIMAL_TUBULE_TRANSPORT	0.13658419	44	1.732e-03	4.226e-01	ATP6V004:43 SLC36A2:135 CA2:152 SLC1A1:216 CA4:253 ABCB1:397
KEGG_SULFUR_METABOLISM	0.30192728	9	1.710e-03	4.226e-01	SULT1A2:97 SULT1A4:370 SULT2B1:1030 SULT1A3:1405 SULT1E1:1680 CHST13:2118
KEGG_TYPE_1_DIABETES_MELLITUS	0.18085786	25	1.753e-03	4.226e-01	CD86:34 TNF:87 HLA-DOB:137 IL1A:608 CD80:787 IFNG:1016
MIKKELSEN_MCV6_HCP_WITH_H3K27ME3	-0.04988831	333	1.866e-03	4.316e-01	CHTRN2:6 SHE4:0 CAMK2B:46 HTRID:55 KCNH2:81 YBX2:87
REACTOME_ARACHIDONIC_ACID_METABOLISM	0.14099083	40	2.044e-03	4.564e-01	PTGR1:96 HPGDS:136 GGT1:470 GPX4:500 CYP1A1:528 GGT5:642
REACTOME_CYTOCHROME_P450_ARRANGED_BY_SUB	0.14731339	35	2.573e-03	3.575e-01	CYP2F1:176 CYP11A1:201 CYP51A1:375 CYP21A2:412 CYP27A1:499 CYP1A1:528
KEGG_GRAFT_VERSUS_HOST_DISEASE	0.21133524	17	2.559e-03	5.375e-01	CD86:34 TNF:87 HLA-DOB:137 IL1A:608 CD80:787 IFNG:1016
PID_NFAT_TFPATHWAY	0.14661319	35	2.697e-03	5.458e-01	TNF:87 FOXK3:324 DGKA:353 EGFR:346 IL4:679 CTLA4:865
REACTOME_MEIOTIC_RECOMBINATION	-0.15454536	31	2.912e-03	5.714e-01	MRE11:406 ATM:434 H2A2:2626 MLH1:818 H3-4831 H2BC9:1058
REACTOME_TNFS_BIND_THEIR_PHYSIOLOGICAL_R	0.17270343	23	3.271e-03	6.186e-01	TNFSF15:112 TNFRSF13B:400 TNFSF14:613 TNFSF8:708 TNFRSF9:836 FASLG:1216
PYEOM_HPV_POSITIVE_TUMORS_UP	0.09915087	73	3.436e-03	6.186e-01	TASOR:41 TAF7:154 UHRF1:155 TRIM56:411 GLS2:427
KEGG_ALLOGRAFT_REJECTION	0.18445481	21	3.439e-03	6.186e-01	CD86:34 TNF:87 HLA-DOB:137 IL4:679 CD80:787 IFNG:1016
LASTOWSKA_NEUROBLASTOMA_COPY_NUMBER_UP	-0.07130891	138	3.908e-03	6.214e-01	MINK1:51 FTS3:58 LRRC59:71 NEURLA:200 CHRN1B:265 PAHB:270
BIOCARTA_HBX_PATHWAY	-0.36844491	5	4.327e-03	6.214e-01	SHC1:353 CREB1:565 LAMTOR5:675 SOS1:955 PTK2B:5819 NA
IVANOVA_HEMATOPOIESIS_MATURE_CELL	0.05934998	203	3.673e-03	6.214e-01	ST3GAL5:84 IL33:127 GCNT1:129 TSPN8:167 TM4SF4:195 COL1A1:209
REACTOME_TP53_REGULATES_TRANSCRIPTION_OF	-0.16105691	26	4.487e-03	6.214e-01	TP53BP2:38 ATM:434 CASP2:465 AIFM2:528 TP73:610 PERP:828

DisGeNET Top pathways by non-permutation

Geneset	stat	num.genes	pval	p.adj	gene.vals
Dermatitis	0.10566782	138	1.950e-05	6.349e-02	TNF:87 ITGAX:104 TNFSF15:112 IL33:127 HPGDS:136 CASP14:151
Leprosy	0.13214844	88	1.908e-05	6.349e-02	TNF:87 TNFSF15:112 HLA-DOB:137 FOXP3:324 IL18RAP:416 CD163:524
Malignant neoplasm of colon stage IV	0.23651636	29	1.347e-05	6.349e-02	PROM1:22 TNFSF10:103 TPX2:286 SMP1:606 FBLN1:633 VEGFA:756
Hepatitis C	0.05525428	498	3.112e-05	7.599e-02	AFP:1 CLU:17 GTG1C3:29 STAT2:38 IFNAR1:58 IL1R2:66
Severe Dengue	0.19085259	38	4.749e-05	9.275e-02	RBFOX1:36 TNF:87 TL7:238 VEGFA:756 CTLA4:865 CXCL8:1051
Cystic Fibrosis	0.06059623	373	6.930e-05	9.668e-02	AFP:1 BPIFB1:64 DNMT3B:31 CELA3B:42 TRPV5:55 TNF:87
Systemic Scleroderma	0.06286578	347	6.705e-05	9.668e-02	PRSS55:2 IFNAR1:58 ENG:64 TNF:87 TNFSF10:103 DNMT1:113
Psoriasis	0.05100856	522	8.434e-05	1.030e-01	IL1R1L:28 STAT2:38 TNF:87 TNFSF10:103 ITGAX:104 TNFSF15:112
Balkan Nephropathy	0.35334846	10	1.094e-04	1.188e-01	TNF:87 GSTK1:235 ABCB1:397 ALB:530 SLC06A1:599 IL1A:608
Goldenhair Syndrome	-0.36617357	9	1.427e-04	1.393e-01	TSC1:59 FRAS1:279 TBC1D20:311 FGF8:316 GRIP1:538 CHD7:768
Hepatitis	0.07592066	201	2.231e-04	1.676e-01	AFP:1 PROM1:22 SLC01B1:82 TNF:87 TNFSF10:103 AKR1D1:117
IGA Glomerulonephritis	0.08278435	169	2.192e-04	1.676e-01	CD86:34 TNF:87 ITGAX:104 MAN2B1:106 HORMAD2:118 FCGRT:149
Inflammatory dermatosis	0.10099228	115	1.913e-04	1.676e-01	TNF:87 ITGAX:104 TNFSF15:112 IL33:127 HPGDS:136 CASP14:151
Renal salt wasting	0.28288815	14	2.484e-04	1.733e-01	CYP11A1:201 CYP21A2:412 SCNN1A:465 CLCNKA:664 SCNN1G:1157 HSD3B2:1180
Kidney Failure, Chronic	0.05924546	310	3.741e-04	1.857e-01	TRPV1:13 CLU:17 CHDH:77 TNF:87 HSD11B1:107 MSRI:115
Acquired aplastic anemia	0.17905557	34	3.056e-04	1.857e-01	TNF:87 HPGDS:136 GSTK1:235 FOXK3:324 IFNAR1:58 SLC06A1:599
Anemia	0.05613830	344	3.926e-04	1.857e-01	DNMT3B:31 ATP2B4:33 IFNAR1:58 ENG:64 TNF:87 TNFSF10:103
Helminthiasis	0.23432224	20	2.875e-04	1.857e-01	PRSS55:2 ITGAX:104 IL33:127 AGXT:186 FOXK3:324 IL4:679
Nonalcoholic Steatohepatitis	0.08576194	146	3.657e-04	1.857e-01	DNMT3B:31 TNF:87 TNFSF10:103 HSD11B1:107 DNMT1:113 IL33:127
Polymorphous light eruption	0.39168099	7	3.327e-04	1.857e-01	TNF:87 HPGDS:136 GSTK1:235 SLC06A1:599 IL1A:608 ICAM1:1079
Rare Diseases	0.26410762	15	3.992e-04	1.857e-01	DNMT3B:31 AGXT:186 FGFR1:48